A Distributional Semantic Model of Comprehension Plausibility in Persian Compound Nouns

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Introduction and Questions

- The acceptability of word combinations is not determined by morpho-syntactic well-formedness alone but also by plausibility—how naturally concepts can be integrated. In Persian, noun–noun compounding is one of the most productive word-formation processes; however, computational modelling of its plausibility has not been addressed. Distributional Semantics (Harris, 1954; Firth, 1957) laid the foundation for modeling meaning from co-occurrence, later developed into Latent Semantic Analysis (Landauer & Dumais, 1997) and extended by compositional approaches (Baroni, Bernardi & Zamparelli, 2014). These frameworks provide tools to estimate semantic relatedness and predict how compounds are interpreted.

- **Günther & Marelli (2016):** Distributional models predicted English speakers' judgments of novel compounds, shaped by head similarity, modifier similarity, and their interaction.

- **Günther, Petilli & Marelli (2020):** Combined linguistic cues with perceptual features (e.g., shape, color, function), showing that grounding models in such knowledge improves compound processing predictions.

- **(Q1)** Can a computational model predict the plausibility of Persian noun–noun compounds?
- **(Q2)** Which semantic relations make compounds more or less plausible?
- **(Q3)** How closely do model predictions align with human judgments?

Methods

- **Corpus & Semantic Space:**
  - ➢Hamshahri Corpus (1M+ tokens)
  - ➢250 frequent nouns, adjectives, adverbs
  - ➢ Building a semantic space with co-occurrence, positional info, POS filtering, stop word removal, and implementing this Formula:

$$C_{ti} = \sum_{k=1}^{s} \left(\sum_{j=1}^{r} \frac{1}{\left(P_{w_t} - P_{w_i}\right)^2} \right)$$

- **Compound Nouns:**
  - ➢400 attested noun–noun compounds (dictionaries)
  - ➢100 novel compounds created; 25 selected for testing

- **Human Evaluation:**
  - ➢82 native speakers
  - ➢5-point Likert scale (1 = implausible, 5 = plausible)
  - ➢Human ratings used as reference for model evaluation

- **Modeling:**
  - ➢Compounds → semantic vectors
  - ➢Neural network (2 hidden layers) trained on 265 attested compounds (proof of concept)
  - ➢Predictions compared directly with human ratings

Results

Figure 1. An example of a 5-dimensional semantic space for 4 words extracted from the designed corpus (translated to English)

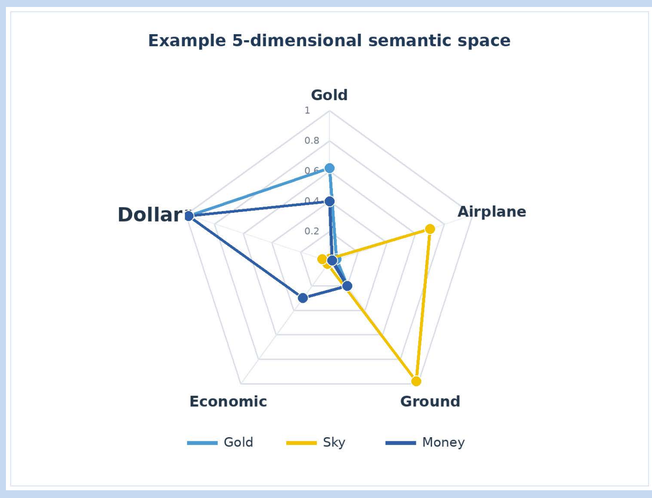
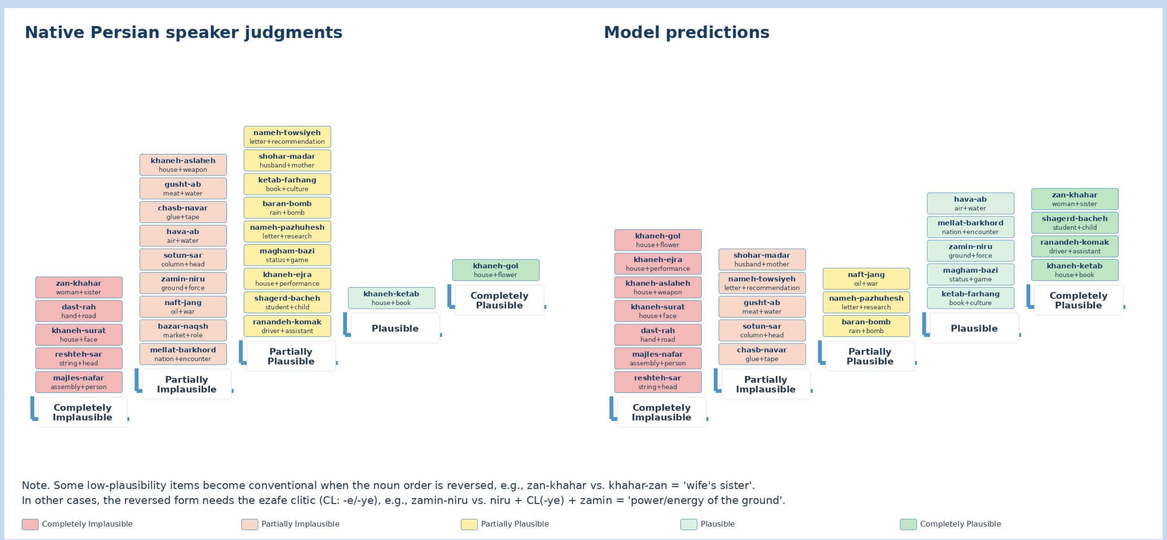


Figure 2. Compound categorization by plausibility level (native Persian speaker judgments, left; model predictions, right)



Interpretation and Outlook

Interpretation:

- In this study, distributional models predicted Persian speakers' plausibility judgments of novel compounds with 77% agreement.
- Plausibility of compounds in Persian is shaped by head–modifier relations and semantic proximity → plausibility depends on conceptual integration.

Further Work:

- Expanding Corpus size and diversity
- Using Diachronic corpora
- Training data enrichment
- Improving the semantic model (e.g., contextual embeddings from BERT)
- Considering Complex predicates

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